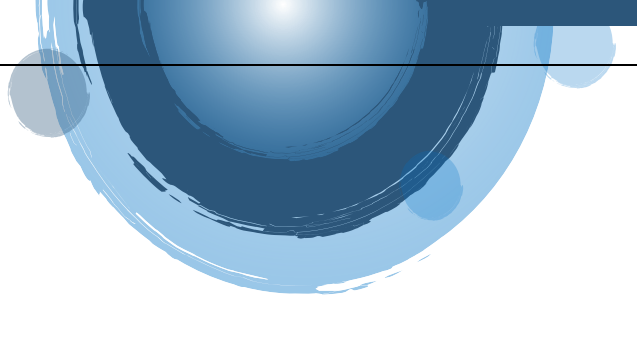




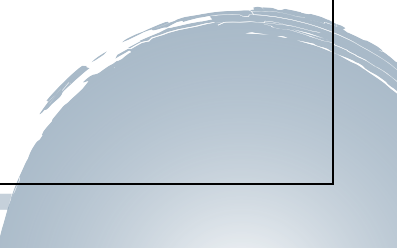
Cyber Security Internship Report

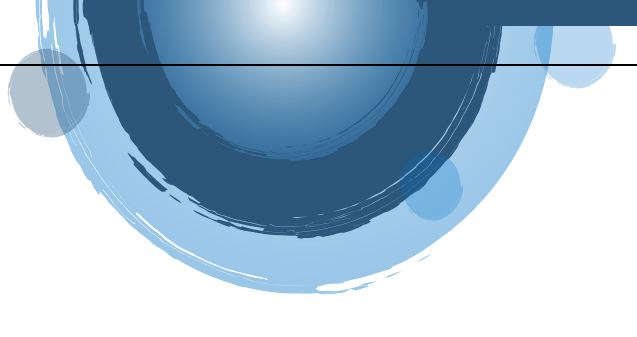
Cyber Security Internship – Task 12

Log Monitoring & Analysis Report

Submitted by: Hetal Antala

Date : 03 / 02 / 2026



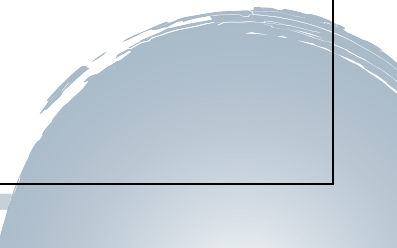


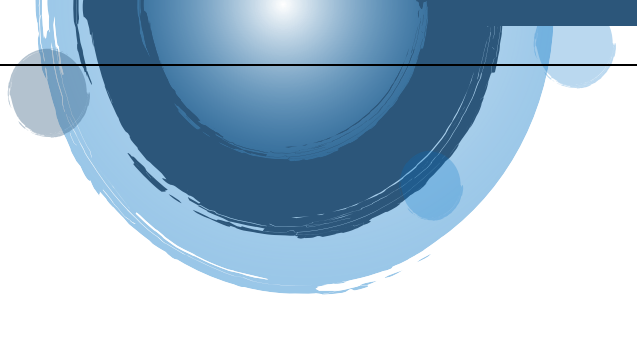
1. Introduction

Logs play a crucial role in cybersecurity for monitoring system activities and detecting suspicious behavior. System logs provide detailed records of authentication attempts, system changes, and user actions. By analyzing logs, security analysts can identify attacks such as brute-force attempts, unauthorized access, and anomalies.

This experiment focuses on monitoring Linux authentication logs using WSL and detecting security incidents using manual analysis and SIEM concepts.

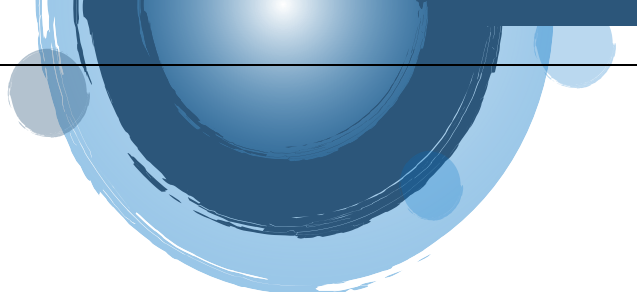
2. Objectives

- Understand different types of system logs
 - Monitor authentication logs
- 

- 
- Identify failed login attempts
 - Detect suspicious activities
 - Learn SIEM basics
 - Develop incident detection skills
-

3. Tools Used

Tool	Purpose
WSL Ubuntu	Linux environment
rsyslog	Log generation
grep/awk/tail	Log analysis
Splunk Free	SIEM (optional)



4. Log Types Analyzed

- Authentication logs (/var/log/auth.log)
 - SSH login attempts
 - Failed passwords
 - Successful logins
 - Sudo access logs
-

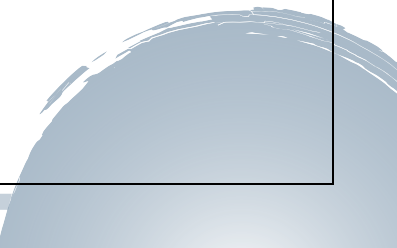
5. Procedure

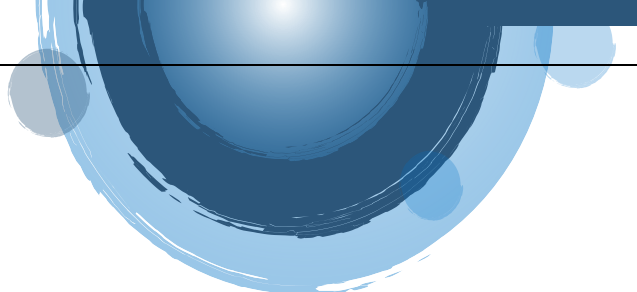
Step 1 – Start services

`sudo service rsyslog start`

`sudo service ssh start`

Step 2 – Generate events

- Incorrect passwords
 - Invalid users
 - SSH attempts
- 



Step 3 – Monitor logs

```
tail -f /var/log/auth.log
```

Step 4 – Analyze logs

```
grep "Failed password" /var/log/auth.log
```

```
grep "Accepted password" /var/log/auth.log
```

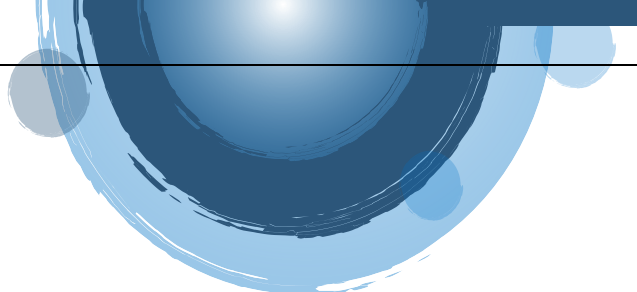
6. Observations

During testing, several authentication events were recorded:

- Multiple failed login attempts
- Invalid users trying to authenticate
- Repeated password failures
- Normal successful login events

Example log entry:

Failed password for invalid user test from
127.0.0.1



7. Findings

- Detected repeated failed login attempts
 - Identified suspicious authentication behavior
 - Possible brute-force attack pattern observed
 - Logs successfully captured all activities
-

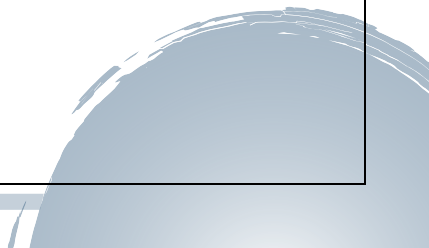
8. Incident Detection

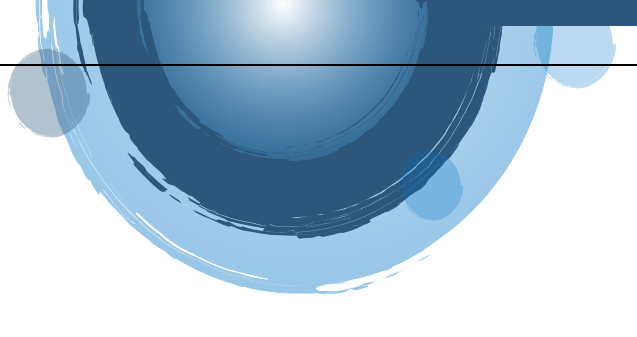
Based on log analysis:

- High number of failures in short time
- Same source attempting multiple logins

This indicates:

👉 Potential brute-force attack





9. SIEM Concept

A SIEM system collects logs from multiple sources, correlates events, and generates alerts. Tools like Splunk help automate detection and improve response time.

10. Conclusion

This task helped in understanding how logs are used to detect cyber threats. By analyzing authentication logs, suspicious activities were successfully identified. The experiment improved practical knowledge of log monitoring, event correlation, and incident detection, which are essential skills for SOC analysts.

